



Figure 4: LSP placement in kernel source

mach-types.h. The macros defined by *mach-types.h* are used by the rest of the kernel code. For example, the machine ID for Beagle Board is 1546, and this is the number which the second stage bootloader passes to the kernel. For registering the new board for ARM, provide the board details at <http://www.arm.linux.org.uk/developer/machines/?action=new>.



Note: The porting concepts described over here are specific to boards based on the ARM platform and may differ for other architectures.

MACHINE_START macro

One of the steps involved in kernel porting is to define the initialisation functions for the various interfaces on the board, such as serial, Ethernet, Gpio, etc. Once these functions are defined, they need to be linked with the kernel so that it can invoke them during boot-up. For this, the kernel provides the macro `MACHINE_START`. Typically, a `MACHINE_START` macro looks like what's shown below:

```
MACHINE_START(MY_BOARD, "My Board for Demo")
    .atag_offset    = 0x100,
    .init_early     = my_board_early,
    .init_irq       = my_board_irq,
    .init_machine   = my_board_init,
MACHINE_END
```

Let's understand this macro. `MY_BOARD` is machine ID defined in *arch/arm/tools/mach-types*. The second parameter to the macro is a string describing the board. The next few lines specify the various initialisation functions, which the kernel has to invoke during boot-up. These include the following:

.atag_offset: Defines the offset in RAM, where the boot parameters will be placed. These parameters are passed from the second stage bootloader, such as u-boot.

my_board_early: Calls the SOC initialisation functions.

This function will be defined by the SOC vendor, if the kernel is ported for it.

my_board_irq: Initialisation related to interrupts is done over here.

my_board_init: All the board-specific initialisation is

done here. This function should be defined during the board porting. This includes things such as setting up the pin multiplexing, initialisation of the serial console, initialisation of RAM, initialisation of Ethernet, USB and so on.

`MACHINE_END` ends the macro. This macro is defined in *arch/arm/include/asm/mach/arch.h*.

How to begin with porting

The most common and recommended way to begin with porting is to start with some reference board, which closely resembles yours. So, if you are porting for a board based on OMAP3 architecture, take Beagle Board as a reference. Also, for porting, you should understand the system very well. Depending on the features available on your board, configure the kernel accordingly. To start with, just enable the minimal set of features required to boot the kernel. This may include but not be limited to initialisation of RAM, Gpio subsystems, serial interfaces, and filesystems drivers for mounting the root filesystem. Once the kernel boots up with the minimal configuration, start adding the new features, as required.

So, let's summarise the steps involved in porting:

1. The first step is to register the machine with the kernel maintainer and get the unique ID for your board. While this is not necessary to begin with porting, it needs to be done eventually, if patches are to be submitted to the mainline. Place the machine ID in *arch/arm/tools/mach-types*.
2. Create the board-specific file '*board-<board_name>*' at *arch/arm/mach-<soc>* and define the `MACHINE_START` for the new board. For example, the board-specific file for the Panda Board resides at *arch/arm/mach-omap2/board-omap4panda.c*.
3. Update the Kconfig file at *arch/arm/mach-<soc>* to add an entry for the new board as shown below:

```
config MACH_MY_BOARD
    bool "My Board for Demo"
    depends on ARCH_OMAP3
    default y
```

4. Update the corresponding makefile, so that the board-specific file gets compiled. This is shown below:

```
obj-$(CONFIG_MACH_MY_BOARD) += board-my_board.o
```

5. Create a default configuration file for the new board. To begin with, take any *.config* file as a starting point and customise it for the new board. Place the working *.config* file at *arch/arm/configs/my_board_defconfig*. 

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